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# Autonomous Navigation of a Differential-Drive Boat on Guerlédan Lake

Project report: compass calibration, GPS guidance, digital twin and lake trials


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**Presentation slides**
vamshidharre.github.io/BEST_Breast/navyseals_presentation.html

**Source code (GitHub)**
github.com/vamshidharre/BEST_Breast

**Boat code, logs & KML**
github.com/vamshidharre/BEST_Breast/tree/main/NavySeals


ABSTRACT

We programmed a DD-Boat, a small autonomous surface vessel steered only by the speed difference of two thrusters, to navigate GPS waypoints on Guerlédan Lake. The work covered magnetometer calibration, projection of GPS fixes to a local metric frame, a proportional heading controller with a spin-in-place mode, and a digital twin used to test the real mission scripts before deployment. On the lake, the boat completed a 15 m box mission (4/4 waypoints, 111 s, loop closed within 2.8 m) and a high-speed sprint in which it reached a buoy 166 m away within 3.7 m. The return leg was cut 84.6 m before the finish when Wi-Fi dropped. We analyse 3,476 telemetry samples, compare them with the simulation, and list the changes these results call for.


## 1 Introduction and objectives

A DD-Boat has no rudder: turning comes from running the left and right motors at different speeds. Its only navigation sensors are a 9-axis IMU, used as a compass, and a GPS receiver giving about one fix per second. All control runs on board. The laptop is only used to start missions and download logs over the lake Wi-Fi. We set three missions of increasing difficulty:

| MISSION | GOAL | SCRIPT | STATUS |
|---|---|---|---|
| **1 · 15 m box** | Visit four waypoints on a 15 m square and return to the start | `waypoint_mission.py` | **Completed on the lake** |
| **2 · Sprint** | Dock → A → B → A at full speed between two buoys 165.7 m apart | `waypoint_mission.py` | **B reached; return cut by Wi-Fi drop** |
| **3 · Regatta** | Cross two gates in the scoring direction as often as possible in 5 min | `regatta_loop.py` | **Built and validated in simulation** |

## 2 Platform

- **Propulsion:** two brushless thrusters driven by an Arduino Mega 2560; no rudder.
- **Computer:** Raspberry Pi 3, Raspbian Stretch, Python 3.5.3 (no f-strings, no SciPy: NumPy only).
- **Sensors:** Pololu 9-DOF IMU (LIS3MDL magnetometer, LSM6DS33 accel/gyro) and a serial NMEA GPS.
- **Operations:** SSH over the lake Wi-Fi; CSV logs at about 9 Hz and KML tracks for Google Earth.
- **Vessels:** DD-Boat 11 for the lake trials, then DD-Boat 18, where the compass axis problem of §3.2 appeared.



# 3 Method

## 3.1 Boat model

We use the nautical convention (θ = 0 north, clockwise positive; x east, y north). The digital twin integrates this model; $u_L$, $u_R$ are motor PWM commands and ($w_x$, $w_y$) a constant wind/current drift.

$$\dot{x} = v\sin\theta + w_x \qquad \dot{y} = v\cos\theta + w_y \qquad \dot{\theta} = k_1(u_L - u_R) \qquad \dot{v} = k_2(u_L + u_R)/2 - c\,v|v|$$

## 3.2 Compass calibration and the axis fix

Nearby iron (screws, motors, battery) shifts and squashes the magnetic field, so a full turn of the boat traces an off-centre ellipse instead of a circle. From one slow 360° turn we fit the offset $b$ and the matrix $A$, then correct each reading:

$$m = A^{-1}\cdot(m_{raw} - b)$$
$$\theta = \operatorname{atan2}(-m_y, m_x) \quad \text{heading, marine convention}$$

**Field problem.** After moving our code to Boat 18, North and South were right but East and West were swapped. The magnetometer's Y axis points the opposite way to the nautical frame. Negating $m_y$ fixed it in `calib.py`, `waypoint_mission.py` and `regatta_loop.py`. Calibration is specific to each hull.

## 3.3 GPS to local metres

Over a few hundred metres the Earth is flat enough for an equirectangular projection around a fixed origin ($lat_o$, $lon_o$) = (48.199111° N, 3.014930° W), with R = 6 371 km:

$$x = R\cdot\cos(lat_o)\cdot(lon - lon_o) \quad \text{east}$$
$$y = R\cdot(lat - lat_o) \quad \text{north}$$

| BUOY | LATITUDE, LONGITUDE | X, Y (M) |
|---|---|---|
| **A** shore | 48.198863, -3.015070 | -10.4, -27.6 |
| **B** lake | 48.198259, -3.017114 | -161.9, -94.7 |

A → B is 165.7 m on a bearing of 246.1° (return 66.1°). To avoid acting on stale data, the GPS serial buffer is drained every cycle and only the newest fix is used.

## 3.4 Guidance and control

The bearing to the active waypoint is compared with the compass heading; the error is wrapped to ±180° with a sawtooth function and fed to a proportional law on the two motors:

$$\theta^* = \operatorname{atan2}(x_t - x,\ y_t - y) \qquad e = \text{sawtooth}(\theta^* - \theta) \qquad u_L = v_0 + K_p\cdot e, \quad u_R = v_0 - K_p\cdot e$$

Both commands are clipped to [0, 115]. When |e| exceeds 40° the boat switches to a **spin-in-place** turn ($u_L = -u_R$, ±75 in the box run and ±95 in the sprint) and stays in it until |e| < 20°. The hysteresis stops it from switching back and forth. A waypoint is reached when the boat is within the arrival radius, or when it has come within 3.5 m and crossed the line through the waypoint perpendicular to the leg ($\langle b - a,\ m - b\rangle > 0$). This catches near misses without circling back. Box settings: $v_0$ = 70, $K_p$ = 28, radius 2.5 m. Sprint settings: $v_0$ = 100, $K_p$ = 35, radius 3.5 m.

## 3.5 Digital twin

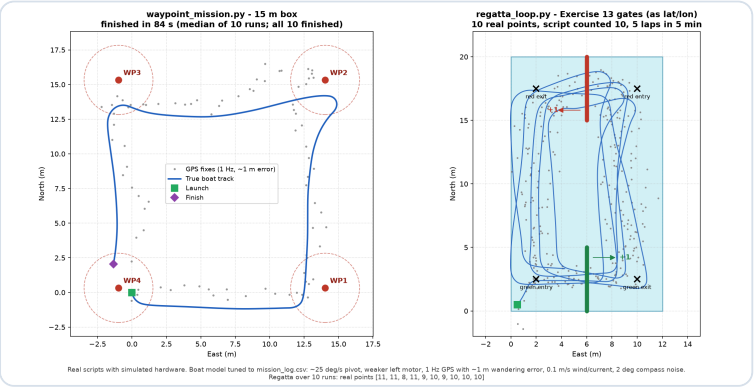



**Fig. 1.** The unmodified mission scripts driving a simulated boat (1 Hz GPS with ~1 m error, 2° compass noise, 0.1 m/s drift, weaker left motor). Left: 15 m box. Right: regatta gates.

Following A. Morvan's method, we swapped the hardware drivers for the model above with realistic sensor noise and ran the **real** mission scripts against it. In 10 runs every box mission finished (median 84 s). The regatta loop scored 8–11 points per 5-minute race, its own count matching the true gate crossings. On the real buoy coordinates, the twin predicted about 140 s for A → B.

# 4 Field results

## 4.1 Mission 1: 15 m box (DD-Boat 11)

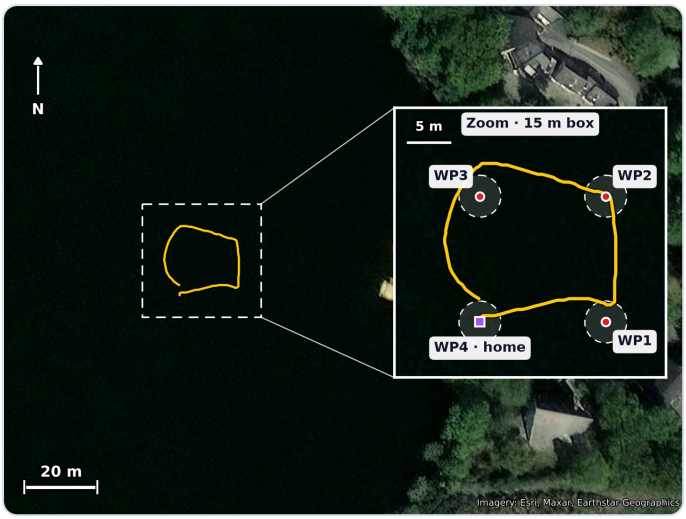



**Fig. 2.** Logged GPS track of the box mission on satellite imagery, with a zoom on the 15 m square. Dashed circles: 2.5 m arrival radius. Data: box_mission_track.kml, box_mission_log.csv.

**4/4**
waypoints reached

**111 s**
total time (sim: 84 s)

**2.8 m**
finish to start distance

**0.59 m/s**
mean speed, 65 m path

| LEG | TIME (S) | CLOSEST (M) |
|---|---|---|
| Start → WP1 (15 m east) | 26.7 | 2.5 |
| WP1 → WP2 | 19.6 | 2.6 |
| WP2 → WP3 | 30.7 | 3.4 |
| WP3 → WP4 (home) | 32.5 | 2.8 |

WP3 was validated by the line-crossing rule (closest pass 3.4 m, outside the 2.5 m circle). Corners are rounded because the boat turns while still moving.

## 4.2 Mission 2: Dock → A → B → A sprint

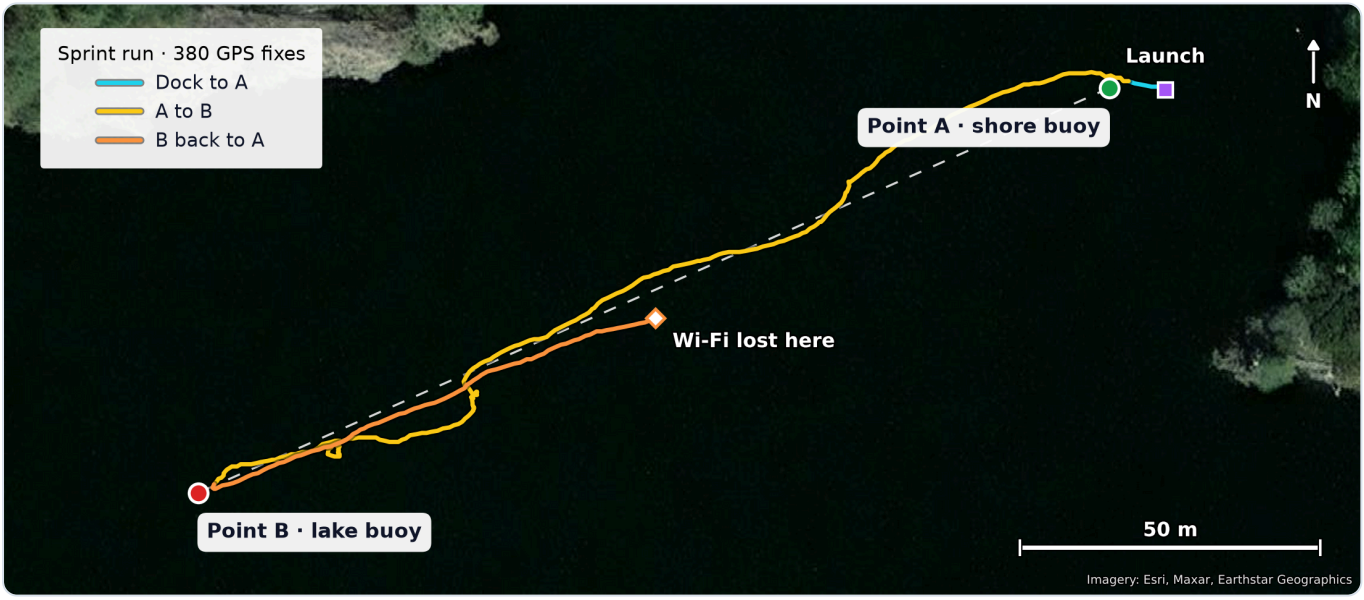



**Fig. 3.** Sprint track from 380 GPS fixes (mission_track.kml), coloured by leg. White dashes: the ideal A–B line. Imagery © Esri, Maxar, Earthstar Geographics.

| LEG | STRAIGHT-LINE | TIME | SPEED MADE GOOD | SPIN TURNS | OUTCOME |
|---|---|---|---|---|---|
| Dock → A | 9.4 m | 11.7 s | 0.44 m/s | 1 | **Reached** |
| A → B | 165.7 m | 280.9 s | 0.59 m/s | 72 (81 s) | **Reached B within 3.7 m** |
| B → A | 165.7 m | 88.2 s | 0.89 m/s | 10 | **Stopped 84.6 m from A** |
| **Total** | | **381 s** | | | 280 m travelled · 3,476 log rows |

**Why the run stopped.** The mission was started directly in the SSH session. When Wi-Fi dropped on the return leg, the script and its log stopped with it, 84.6 m before A. The boat was still on course at near full power (u ≈ 101/98). Running missions inside `screen -S sprint` keeps them alive through a disconnect. The planned re-run was cancelled because of heavy wind and rain.

### 4.3 Telemetry analysis

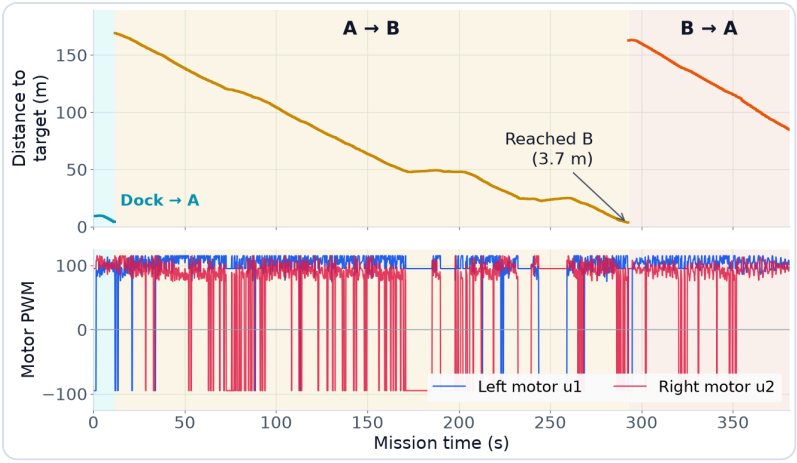



**Fig. 4.** Sprint telemetry (**mission_log.csv**). Top: distance to the active waypoint. Bottom: motor commands; drops to −95 mark spin-in-place turns.

The outbound leg was the weak point. The heading error crossed the 40° threshold 72 times, and the boat spent 81 s (29% of the leg) spinning in place. At 0.59 m/s made good, the leg took 281 s, twice the 140 s the twin predicted. The two plateaus in the distance curve, around 170–200 s and 230–260 s, are where the boat stalled, spinning and drifting. On the way back, the same controller made only 10 spin turns and held 0.89 m/s with both motors near full power. The code and gains were identical in both directions, which suggests the outbound heading was fighting the wind and waves rather than a software fault.

## 5 Reliability and safety

- **GPS watchdog:** no fix for 3 s sets both motors to zero.
- **Track autosave:** the KML is rewritten every 5 s, so a power cut loses almost nothing.
- **Emergency stop:** `stop.py` forces (0, 0) immediately.
- **Auto download:** a laptop script waits for the boat to reconnect, then copies the KML and CSV.
- **Detached sessions** (lesson from the sprint): starting missions inside `screen` keeps them running through a Wi-Fi drop.

## 6 Discussion

**Simulation vs lake.** The twin was right about behaviour (all waypoints reached, correct gate logic) but optimistic about time: 84 s vs 111 s for the box, 140 s vs 281 s for A → B. Its wind and wave model is too gentle, and spin turns cost more on real water.

**Sensors.** One sign error swapped East and West on a new boat: each hull needs its own calibration and a four-point compass check.

**Operations.** The sprint was limited by how the script was launched, not by the controller.

## 7 Conclusion and future work

We built and tested a full GPS navigation stack for the DD-Boat: compass calibration, local projection, a hysteretic proportional controller, line-crossing waypoint validation, safety watchdogs and logging. It completed the 15 m box on the lake and reached the far buoy of the 166 m sprint within 3.7 m. Next steps: (i) re-run the sprint inside `screen` to record the full A → B → A time; (ii) cut spin turns on long legs by tuning the 40°/20° thresholds and adding cross-track or wind-drift correction; (iii) race the regatta loop on the water; (iv) feed the measured speed and turn rate back into the twin ($k_1$, $k_2$) so its time predictions match the lake.

## 8 Reproducibility

Every figure is generated from the logged data in the repository. `python make_slide_images.py` rebuilds Fig. 2–4 from `NavySeals/box_mission_log.csv` and `NavySeals/mission_log.csv`; `python report/build_report.py` rebuilds this PDF. On the boat (Python 3.5.3), missions are started with `screen -S sprint python3 waypoint_mission.py`, and `python3 stop.py` is the emergency stop.

## 9 Resources and references


1. Slides: vamshidharre.github.io/BEST_Breast/navyseals_presentation.html · Code: github.com/vamshidharre/BEST_Breast
2. L. Jaulin, *DD-Boat introductory course* (best_ddboat.pdf), ENSTA Bretagne, BEST 2026: calibration, control law and finish-line test.
3. L. Jaulin, *Mobile robotics / drones* textbook, ch. 4 (DD-Boats), p. 28: local GPS projection.
4. A. Morvan, *Digital twins and observers*, lecture slides, ENSTA Bretagne, 2026.
5. Esri World Imagery basemap (Esri, Maxar, Earthstar Geographics), used for Fig. 2–3.